

Terraform Configuration Documentation

1. provider.tf

The `provider.tf` file specifies the cloud provider (in this case, AWS) and the region where the resources will be provisioned. It uses the `region` variable defined in `variables.tf`.

```
provider "aws" {  
  region = var.region  
}
```

2. variables.tf

The `variables.tf` file contains input variables used throughout the Terraform configuration. These variables allow customization of infrastructure without modifying the code.

Defined Variables:

```
variable "region" {  
  type = string  
  default = "us-west-2"  
}
```

```
variable "cidr" {  
  description = "The CIDR block for the VPC. Default value is a valid CIDR, but not acceptable by AWS  
and should be overridden"  
  type = string  
  default = "10.0.0.0/16"  
}
```

```
variable "instance_tenancy" {  
  description = "A tenancy option for instances launched into the VPC"  
  type = string  
  default = "default"  
}
```

```
variable "enable_dns_hostnames" {  
  description = "Should be true to enable DNS hostnames in the VPC"  
  type = bool  
  default = true  
}
```

```

variable "enable_dns_support" {
  description = "Should be true to enable DNS support in the VPC"
  type        = bool
  default     = true
}

variable "enable_classiclink" {
  description = "Should be true to enable ClassicLink for the VPC. Only valid in regions and accounts
that support EC2 Classic."
  type        = bool
  default     = false
}

variable "tags" {
  description = "A map of tags to add to all resources"
  type        = map
  default     = {
    Name = "Vpc-custom-demo"
  }
}

```

3. main.tf

The `main.tf` file defines the core infrastructure resources. Below are the components and their roles:

Resources Defined:

```

resource "aws_vpc" "vpc_demo" {
  cidr_block      = var.cidr
  instance_tenancy = var.instance_tenancy
  enable_dns_hostnames = var.enable_dns_hostnames
  enable_dns_support = var.enable_dns_support

  tags = {
    Name = var.tags["Name"]
  }
}

resource "aws_internet_gateway" "gw" {
  vpc_id = aws_vpc.vpc_demo.id

  tags = {
    Name = "internet-gateway-demo"
  }
}

```

```
resource "aws_subnet" "public_1" {
  vpc_id          = aws_vpc.vpc_demo.id
  map_public_ip_on_launch = true
  cidr_block      = "10.0.1.0/24"

  tags = {
    Name = "public_1-demo"
  }
}

resource "aws_route_table" "route-public" {
  vpc_id = aws_vpc.vpc_demo.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.gw.id
  }

  tags = {
    Name = "public-route-table-demo"
  }
}

resource "aws_route_table_association" "public_1" {
  subnet_id    = aws_subnet.public_1.id
  route_table_id = aws_route_table.route-public.id
}
```